

Ubuntu 24.04 + Docker + kubectl + Minikube

Cloud VM uchun toza va tavsiya etilgan o'rnatish ketma-ketligi

1. Update system

```
sudo apt update && sudo apt upgrade -y
```

2. Remove old/conflicting Docker packages

```
sudo apt remove -y \  
  docker.io \  
  docker-compose \  
  docker-compose-v2 \  
  docker-doc \  
  docker-buildx \  
  podman-docker \  
  containerd \  
  runc
```

2.1 Install prerequisites

```
sudo apt install -y ca-certificates curl
```

2.2 Add Docker GPG key

```
sudo install -m 0755 -d /etc/apt/keyrings
```

```
sudo curl -fsSL \  
  https://download.docker.com/linux/ubuntu/gpg \  
  -o /etc/apt/keyrings/docker.asc
```

```
sudo chmod a+r /etc/apt/keyrings/docker.asc
```

2.3 Add Docker repository

```
sudo tee /etc/apt/sources.list.d/docker.sources > /dev/null <<EOF  
Types: deb  
URIs: https://download.docker.com/linux/ubuntu  
Suites: $(. /etc/os-release && echo "${UBUNTU_CODENAME:-$VERSION_CODENAME}")  
Components: stable  
Architectures: $(dpkg --print-architecture)  
Signed-By: /etc/apt/keyrings/docker.asc  
EOF
```

2.4 Install Docker Engine

```
sudo apt update
```

```
sudo apt install -y \  
  docker-ce \  
  docker-ce-cli \  
  containerd.io \  
  docker-buildx-plugin \  
  docker-compose-plugin
```

2.5 Start Docker

```
sudo systemctl enable --now docker  
sudo systemctl status docker --no-pager
```

```
docker --version  
docker compose version
```

2.6 Allow the user to run Docker without sudo

```
sudo usermod -aG docker $USER
newgrp docker

groups
docker ps
docker run hello-world
```

3. Install kubectl

```
curl -LO "https://dl.k8s.io/release/$(curl -L -s https://dl.k8s.io/release/stable.txt)/bin/linux/amd64/kubectl"

sudo install -o root -g root -m 0755 kubectl /usr/local/bin/kubectl

rm -f kubectl

kubectl version --client
```

4. Install Minikube

```
curl -LO https://storage.googleapis.com/minikube/releases/latest/minikube-linux-amd64

sudo install minikube-linux-amd64 /usr/local/bin/minikube

rm -f minikube-linux-amd64

minikube version
```

5. Start Minikube with Docker driver

```
whoami
minikube start --driver=docker
```

6. Verify the cluster

```
minikube status
kubectl get nodes
kubectl get pods -A
```

Important notes

- Minikube'ni root user bilan Docker driver orqali ishga tushirmang. Oddiy user (masalan, ubuntu) bilan ishlatang.
- Docker repository qo'shilgandan keyin albatta sudo apt update bajaring.
- docker ps sudo'siz ishlashi uchun user docker group'iga qo'shilgan bo'lishi kerak.
- docker run hello-world muvaffaqiyatli ishlasa, Docker tayyor hisoblanadi.

Architecture

```
Ubuntu 24.04
├── Docker Engine
│   ├── Container runtime
│   ├── kubectl
│   │   ├── Kubernetes CLI
│   │   └── Minikube
│       ├── Kubernetes cluster
│       └── API Server
```

■■■ Scheduler
■■■ Controller Manager
■■■ CoreDNS